

Practical 6

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Aim: For a given set of relation schemes, create tables and perform the following Simple Queries, Simple Queries with Aggregate functions, Queries with Aggregate functions (group by and having clause).

Source-Code & Output:

Create table Student2(StudId INT,Name varchar(20),Dept varchar(20),Marks INT);
use rishabh;

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' tree with a database named 'rishabh' selected. The main editor window contains the following SQL code:

```
1 • Create table Student2(StudId INT,Name varchar(20),Dept varchar(20),Marks INT);
2 • use rishabh;
3 • INSERT INTO Student2 values(1,"Rishabh Dwivedi","MCA",75);
4 • INSERT INTO Student2 values(2,"Aman Singh","BCA",85);
5 • INSERT INTO Student2 values(3,"Pranit Bhairam","BSC",95);
6 • INSERT INTO Student2 values(4,"Vijay Thakur","CS",88);
7 • INSERT INTO Student2 values(5,"Anant Sahu","HR",99);
8 • INSERT INTO Student2 values(6,"Shubham Dwivedi","MCA",73);
9 • INSERT INTO Student2 values(7,"Raman Singh","BCA",84);
10 • INSERT INTO Student2 values(8,"Pronit Bhairam","BSC",98);
11 • INSERT INTO Student2 values(9,"Jay Thakur","CS",80);
12 • INSERT INTO Student2 values(10,"Raman Sahu","HR",91);
13 • INSERT INTO Student2 values(11,"Rishi Dwivedi","MCA",72);
14 • INSERT INTO Student2 values(12,"Varun Singh","BCA",85);
15 • select * from Student2;
```

The bottom panel shows the 'Output' tab with a table of execution results:

#	Time	Action	Message
11	12:22:41	select max(Marks) as highest_marks from Student2 LIMIT 0, 1000	1 row(s) returned
12	12:22:48	select min(Marks) as lowest_marks from Student2 LIMIT 0, 1000	1 row(s) returned
13	12:22:56	select sum(Marks) as total_marks from Student2 LIMIT 0, 1000	1 row(s) returned
14	12:29:04	drop table Student2	0 row(s) affected
15	12:32:54	Create table Student2(StudId INT,Name varchar(20),Dept varchar(20),Marks INT)	0 row(s) affected
16	12:32:54	use rishabh	0 row(s) affected

```
INSERT INTO Student2 values(1,"Rishabh Dwivedi","MCA",75);
INSERT INTO Student2 values(2,"Aman Singh","BCA",85);
INSERT INTO Student2 values(3,"Pranit Bhairam","BSC",95);
INSERT INTO Student2 values(4,"Vijay Thakur","CS",88);
INSERT INTO Student2 values(5,"Anant Sahu","HR",99);
INSERT INTO Student2 values(6,"Shubham Dwivedi","MCA",73);
INSERT INTO Student2 values(7,"Raman Singh","BCA",84);
INSERT INTO Student2 values(8,"Pronit Bhairam","BSC",98);
INSERT INTO Student2 values(9,"Jay Thakur","CS",80);
INSERT INTO Student2 values(10,"Raman Sahu","HR",91);
INSERT INTO Student2 values(11,"Rishi Dwivedi","MCA",72);
INSERT INTO Student2 values(12,"Varun Singh","BCA",85);
select * from Student2;
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'rishabh' selected. The main editor shows four SQL queries: 1. Create table Student2(StudId INT, Name varchar(20), Dept varchar(20), Marks INT); 2. use rishabh; 3. INSERT INTO Student2 values(1, 'Rishabh Dwivedi', 'MCA', 75); 4. INSERT INTO Student2 values(2, 'Aman Singh', 'BCA', 85);. Below the queries, the 'Result Grid' shows a table with 12 rows and 4 columns: StudId, Name, Dept, and Marks. The data is as follows:

StudId	Name	Dept	Marks
1	Rishabh Dwivedi	MCA	75
2	Aman Singh	BCA	85
3	Pranit Bhairam	BSC	95
4	Vijay Thakur	CS	88
5	Anant Sahu	HR	99
6	Shubham Dwivedi	MCA	73
7	Raman Singh	BCA	84
8	Pranit Bhairam	BSC	98
9	Jay Thakur	CS	80
10	Raman Sahu	HR	91
11	Rishi Dwivedi	MCA	72
12	Varun Singh	BCA	85

select count(*) as total_students from Student2;

The screenshot shows the MySQL Workbench interface with the SQL query: 16 • select count(*) as total_students from Student2;. The 'Result Grid' shows a single row with the column 'total_students' and the value 12.

total_students
12

select avg(Marks) as average_marks from Student2;

The screenshot shows the MySQL Workbench interface with the SQL query: 17 • select avg(Marks) as average_marks from Student2;. The 'Result Grid' shows a single row with the column 'average_marks' and the value 85.4167.

average_marks
85.4167

select max(Marks) as highest_marks from Student2;

The screenshot shows the MySQL Workbench interface with the SQL query: 18 • select max(Marks) as highest_marks from Student2;. The 'Result Grid' shows a single row with the column 'highest_marks' and the value 99.

highest_marks
99

select min(Marks) as lowest_marks from Student2;

The screenshot shows the MySQL Workbench interface with the SQL query: 19 • select min(Marks) as lowest_marks from Student2;. The 'Result Grid' shows a single row with the column 'lowest_marks' and the value 72.

lowest_marks
72

select sum(Marks) as total_marks from Student2;

20 • `select sum(Marks) as total_marks from Student2;`

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	total_marks
▶	1025

select Dept, avg(Marks) as Avg_marks from Student2 group by Dept;

21 • `select Dept, avg(Marks) as Avg_marks from Student2 group by Dept;`

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Dept	Avg_marks
▶	MCA	73.3333
	BCA	84.6667
	BSC	96.5000
	CS	84.0000
	HR	95.0000

select Dept, avg(Marks) as Avg_marks from Student2 group by Dept having avg(Marks)>80;

22 • `select Dept, avg(Marks) as Avg_marks from Student2 group by Dept having avg(Marks)>80;`

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Dept	Avg_marks
▶	BCA	84.6667
	BSC	96.5000
	CS	84.0000
	HR	95.0000